

NHAI HACKATHON 7.0

EdgeLock

Enterprise-Grade Offline-First Face Authentication
& 3D Liveness Detection



TEAM BINARY BRAINS



React Native



Offline-First



On-Device AI

THE PROBLEM

Highway field crews work in zero-connectivity zones

- ✗ **No internet** — Remote highways have zero cellular coverage
- ✗ **Buddy punching** — Proxy attendance with shared credentials
- ✗ **Spoofing attacks** — Photos & videos used to fake identity
- ✗ **Audit gap** — No tamper-proof digital trail of attendance

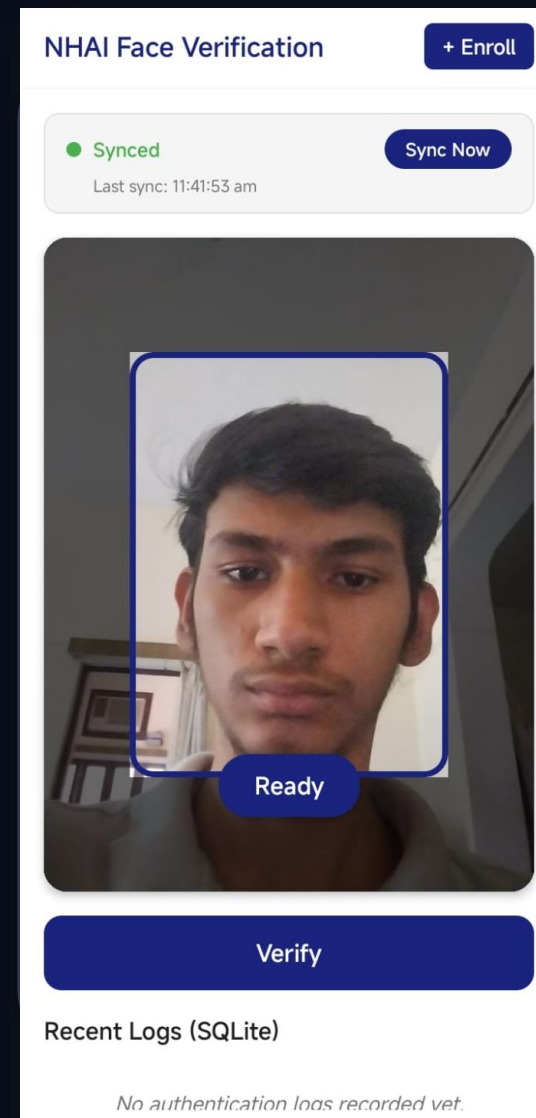


NHAI manages 1,40,000+ km of national highways across India

OUR SOLUTION

100% Offline Face Auth with 3D Liveness Detection

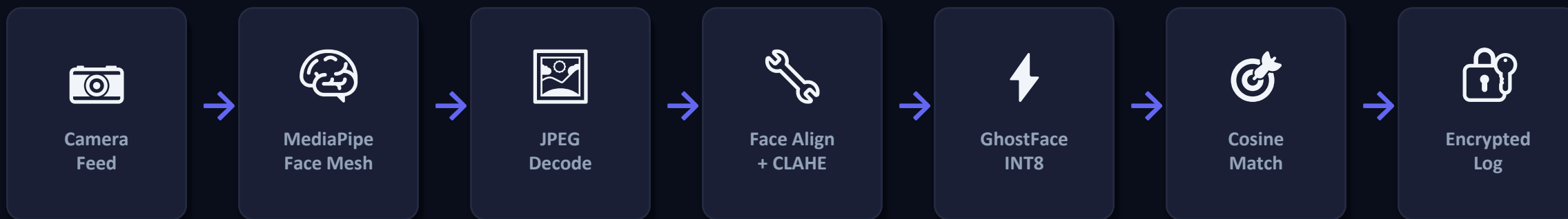
- ✓ **On-device AI** — GhostFaceNet INT8 runs entirely on the phone
- ✓ **3D liveness** — Blink + head turn + depth verification
- ✓ **AES-256 encrypted** — Biometric data secured at rest
- ✓ **Auto-sync** — Logs upload when connectivity returns
- ✓ **GPS-tagged** — Every auth log includes geo-coordinates



ARCHITECTURE

End-to-End Pipeline

Every step runs on-device — zero cloud dependency during authentication



100% On-Device



Real-Time Inference



AES-256 at Rest

TECHNOLOGY

Technical Stack

Battle-tested frameworks, zero cloud dependency at runtime



React Native + Expo

SDK 50 · TypeScript
Strict



TensorFlow Lite JSI

C++ bridge · Sync
inference



GhostFaceNet INT8

1 MB · 512-dim
embeddings



MediaPipe WASM

468 face landmarks



SQLite + AES-256

Encrypted local
database



Hardware Keystore

iOS Keychain · Android
KS



Expo Location

GPS-tagged auth logs



AWS Sync Queue

Zero-loss batch sync

ANTI-SPOOFING

Multi-Factor 3D Liveness Detection

Four independent checks defeat photos, videos, and masks



Eye Aspect Ratio (EAR)

Real-time blink detection — threshold < 0.33



Head Yaw Assessment

Smoothed asymmetry with hysteresis (ON: 0.12, OFF: 0.09)



3D Depth Verification

Z-coordinate std dev separates 3D faces from flat screens



Random Challenge Order

Shuffled per session — unpredictable for attackers

NHAI Face Verification

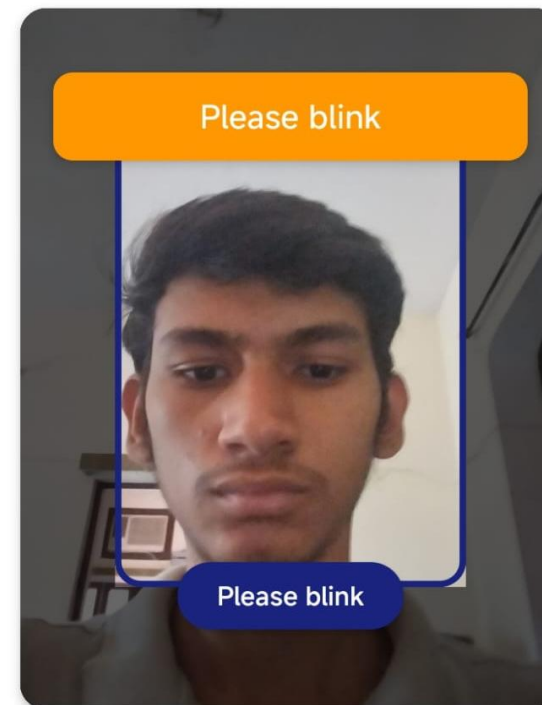
[+ Enroll](#)

● Synced

[Sync Now](#)

Last sync: 8:05:44 pm

Please blink



Please blink

Verify

Recent Logs (SQLite)

No authentication logs recorded yet.

RECOGNITION

GhostFaceNet

INT8 Engine

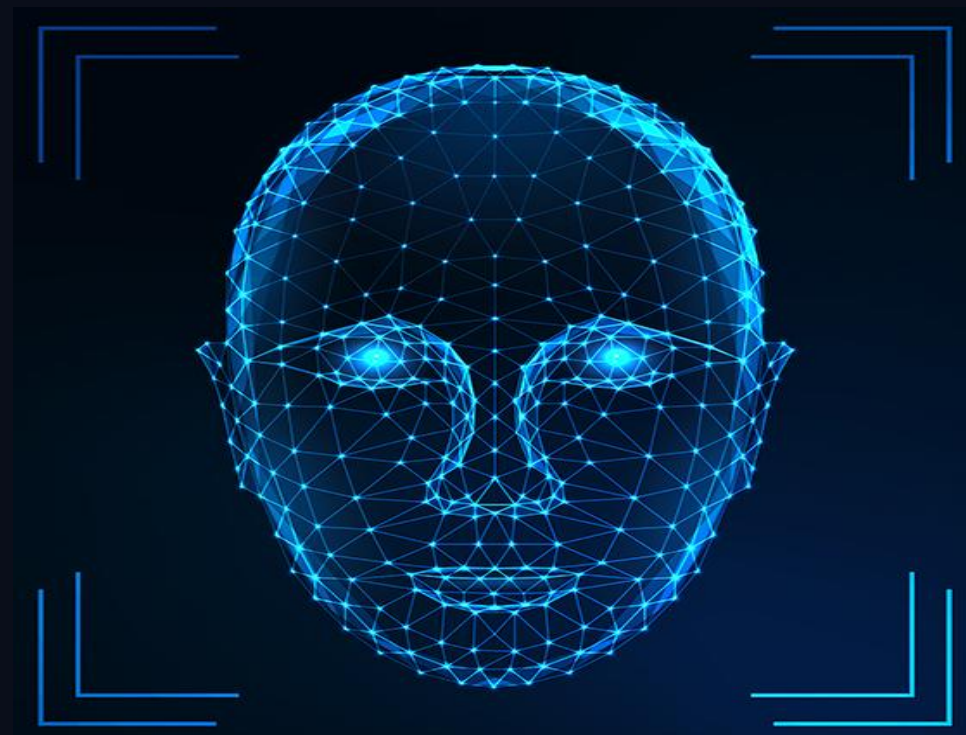
Desktop-grade accuracy in a 1 MB on-device model

```
// Recognition Pipeline
capture → 3-shot average (180ms apart)
preprocess → 5-point face alignment (112×112)
lighting → CLAHE / histogram equalization
inference → model.runSync() via JSI
output → 512-dim L2-normalized embedding
match → cosine similarity vs encrypted DB
```

```
// Thresholds
```

```
single_user: 0.75 | multi_user: 0.84
```

```
high_confidence: 0.91 | margin: 0.05
```



~1 MB

MODEL SIZE

512

DIMENSIONS

SECURITY

Offline-First & Zero-Loss Sync

Data integrity guaranteed — no log is ever lost



Encrypted Storage

- ✓ AES-256-CBC encrypted embeddings
- ✓ Hardware-backed Keychain/Keystore
- ✓ Per-install encryption keys



Zero-Loss Sync

- ✓ Purge only after HTTP 200 + ID match
- ✓ Idempotent server-side processing
- ✓ Auto-sync on network reconnection



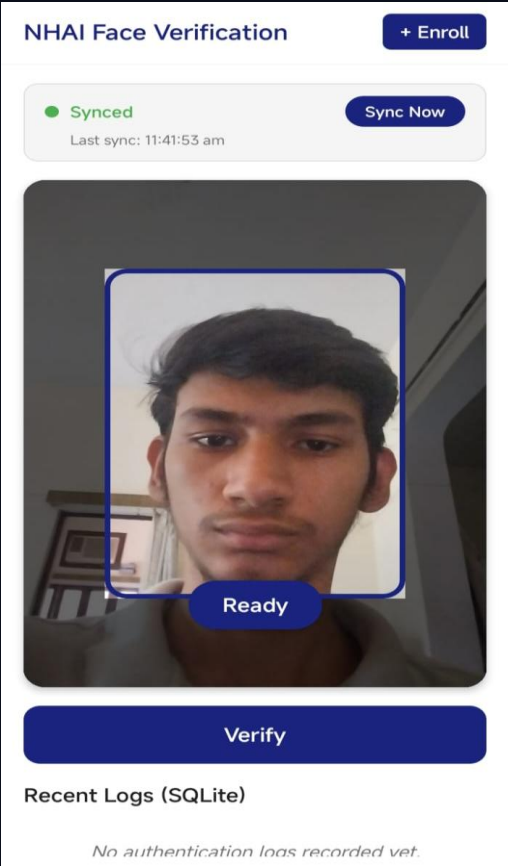
Audit Trail

- ✓ GPS-tagged every auth event
- ✓ Photo thumbnail captured
- ✓ UUID + ISO8601 timestamps

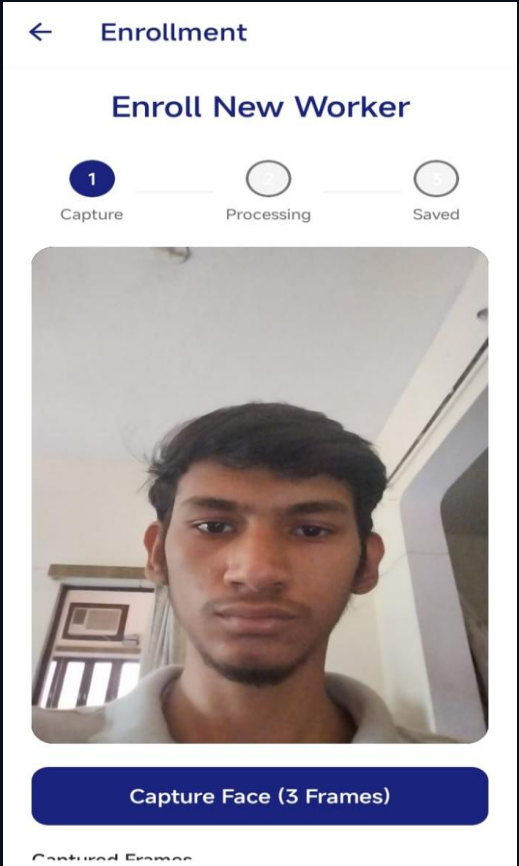
DEMO

App Screens

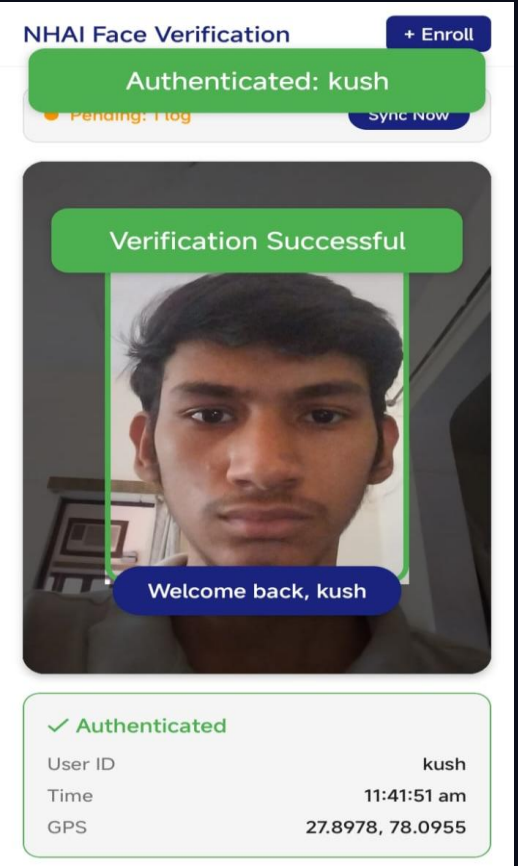
Clean, minimal UI designed for outdoor field operations



Authentication



Face Enrollment



Auth Success

METRICS

By The Numbers

~1 MB

AI MODEL SIZE

512

EMBEDDING DIMS

468

FACE LANDMARKS

0%

CLOUD DEPENDENCY

AES-256

ENCRYPTION

INT8

QUANTIZATION

9

TEST SUITES

0

DATA LOSS RISK

All inference runs on-device via JSI C++ bridge — no network roundtrip required

QUALITY

Tested & Production-Ready

Comprehensive test coverage across every system layer

9 test suites covering:

<code>database.test.ts</code>	→ SQLite CRUD
<code>liveness.test.ts</code>	→ EAR, yaw, depth
<code>preprocessing.test.ts</code>	→ CLAHE, quality
<code>recognition.test.ts</code>	→ Embeddings, cosine
<code>networkSync.test.ts</code>	→ Zero-loss sync
<code>enrollment.test.tsx</code>	→ Multi-capture flow
<code>faceAuth.test.tsx</code>	→ Component render
<code>orchestration.test.tsx</code>	→ Full pipeline
<code>e2e.test.tsx</code>	→ End-to-end sim



Security Guarantees



AES-256 encryption tested



Zero-loss purge verified



Idempotent sync tested



Build & Deploy



EAS cloud builds (no Mac needed)



iOS + Android from Windows



TypeScript strict mode, 0 errors

Thank You

EdgeLock brings enterprise-grade biometric security to India's most remote highways — fully offline.



TEAM BINARY BRAINS

<https://github.com/DeveloperKush/NHAI-Hackathon-7.0>



KUSH SARASWAT



SMARPIT MALIK